

Installing Docker Toolbox

Before installing Docker Toolbox, ensure that your system has 64-bit Windows 7 or higher, and virtualisation is enabled.

The installation steps are as follows:

1. Docker Toolbox is no longer maintained by Docker, so download it from the GitHub URL at <https://github.com/docker/toolbox/releases>. Download the .exe file.
2. Double click the installer for installing Docker Toolbox.
3. If your system already has VirtualBox installed, uncheck it. Also, if VirtualBox is running, stop it before installing Docker Toolbox.
4. Accept all other default settings and install Docker Toolbox. Once it has been successfully installed, it can be verified by the three new icons on your desktop, namely, Docker Quickstart Terminal, Oracle VirtualBox, and Kitematic (Alpha).

Docker Quickstart Terminal

To start using Docker on Windows, click on Docker Quickstart Terminal. This launches the pre-configured Docker Toolbox terminal, which sets up everything on its own. Choose 'Yes' each time it prompts for permission. This terminal runs the bash shell instead of the Windows command prompt, as Docker needs it for running.

Once the setup is completed, a Docker shell will show up, and you can verify by executing the command `docker --version`.

This terminal supports all Docker commands like pulling an image from the Docker Hub and running a container from the pulled image. We can also commit and push an image to Docker Hub.

Let us create a container using Docker Quickstart Terminal with a Jenkins image running in it. The command for this is `docker run -p 8080:8080 -p 50000:50000 jenkins/jenkins:lts`. Figure 1 shows the above Docker command in execution.

```
Shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ docker run -p 8080:8080 -p 50000:50000 jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
3192219afd04: Pull complete
17c160265e75: Pull complete
cc4fe40d0e61: Pull complete
1bfe918b8aa5: Pull complete
dafa1a7c0751: Downloading [----->] 71.31MB/104.2MB
4871da813f56: Download complete
24fa635d5c7a: Download complete
72d7a7deec14: Download complete
f7fc83e75d1c: Downloading [==>] 2.698MB/67.18MB
8f13ab868e19: Waiting
af14f7667e07: Waiting
e9a93a70ada9: Waiting
```

Figure 1: Running a Jenkins image in a container

```
MINGW64/c/Program Files/Docker Toolbox
Username: shriyakabra97
Password:
Error response from daemon: Get https://registry-1.docker.io/v2/: net/http: request canceled while
s)

Shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ docker login
Login with your Docker ID to push and pull images from Docker Hub. If you don't have a Docker ID, head
Username: shriyakabra97
Password:
WARNING! Your password will be stored unencrypted in C:\Users\Shriya Kabra\.docker\config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded

Shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ docker push shriyakabra97/brave_cori:2.0
The push refers to repository [docker.io/shriyakabra97/brave_cori]
3935da11e098: Pushed
3dc9f1af2df1: Mounted from jenkins/jenkins
fcf933db32c5: Mounted from jenkins/jenkins
2b7b60533e4b: Mounted from jenkins/jenkins
7a0557d224d1: Mounted from jenkins/jenkins
7948c3e5790c: Mounted from jenkins/jenkins
2.0: digest: sha256:c53f0f6a67a5ac80d5c5f019d5886cabdb52419f1acbc7d751e07b1efafa115f size: 4720

Shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ -
```

Figure 2: Pushing an image to Docker Hub

```
MINGW64/c/Program Files/Docker Toolbox

shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ uname
MINGW64_NT-10.0

shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ man ls
dash: man: command not found

shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ apt-get update
dash: apt-get: command not found

shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ locate apt-get

shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ sysctl
dash: sysctl: command not found

shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ depmod
dash: depmod: command not found

shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$ insmod
dash: insmod: command not found

shriya Kabra@LAPTOP-ELEUSLI8 MINGW64 /c/Program Files/Docker Toolbox
$
```

Figure 3: Commands not supported by Docker Quickstart Terminal